

Your grade: **100%**

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Next item →

1. What is the primary purpose of using activation functions in a neural network layer?

1 / 1 point

- ☒ To introduce non-linearity into the model
- ☐ To reduce the number of neurons required in each layer
- ☐ To normalize the input data
- ☐ To initialize the weights of the neurons

 **Correct**
Correct! Introducing non-linearity allows the model to learn complex patterns.

2. Why might you choose a ReLU activation function over a sigmoid function for hidden layers?

1 / 1 point

- ☐ ReLU is computationally more complex
- ☐ ReLU is less popular than the sigmoid function
- ☐ ReLU always produces smaller output values
- ☒ ReLU helps mitigate the vanishing gradient problem

 **Correct**
Correct! ReLU helps in mitigating the vanishing gradient problem, which is common with sigmoid functions.

3. What are some of the key parameters that can be set when fitting a multiclass classification model?

1 / 1 point

☒ Validation split

 **Correct**
Correct! Validation split is an important parameter to set when fitting the model.

☒ Epochs

 **Correct**
Correct! The number of epochs is a key parameter when fitting the model.

☐ Learning rate

☒ Batch size

 **Correct**
Correct! Batch size is one of the parameters you can set when fitting the model.

☐ Activation function

4. Which of the following actions can be performed during the process of compiling a multiclass classification model?

1 / 1 point

☒ Defining the loss function

 **Correct**
Correct! Defining the loss function is a crucial step in compiling a model.

☒ Choosing the optimizer

 **Correct**
Correct! Choosing the optimizer is an essential step in the compiling process.

☒ Specifying the evaluation metrics

 **Correct**
Correct! Specifying the evaluation metrics is done during the compiling of the model.

☐ Setting the batch size

☐ Splitting the data into training and validation sets

5. Which function in Keras is used to compile a model before training?

1 / 1 point

- ☐ evaluate()
- ☐ fit()
- ☐ predict()
- ☒ compile()

 **Correct**
Correct! The compile() function is used to configure the learning process before training a model.